

CYBER DEFENDERS SPEAK

ESET RESEARCHERS REVEAL MALWARE'S NEW FRONTIERS

Jayesh Shinde | editor@digit.in

Cybersecurity veterans Righard Zwienenberg and Lukas Stefanko, Senior Research Fellow and Malware Analyst at ESET respectively, offer insights into the threat landscape. With decades of experience, they delve into critical issues from malware trends and IoT vulnerabilities to securing essential infrastructure and encryption.

Q What Android malware scams are rising in India?

A Lukas: In 2025, both globally and in India, the Kaleidoscope (tinyurl.com/kal-malware) malware family that is responsible for unwanted ads and ad fraud. The Kaleidoscope often appears post-installation with a user agreement prompt. Banking malware, notably FatBoyPanel, tries to lure banking credentials and SMS OTPs to access the victim's bank account and their funds.

Q About preventing and stopping attacks?

A Lukas: Two tips to prevent banking malware from intercepting SMS or OTPs. The first one, go to App Info > App Permissions and disable SMS

permissions to stop apps that manipulate them. Second, some apps can intercept software-generated OTPs. Go to Accessibility services and disable for apps you don't trust.

Measures on Android are:

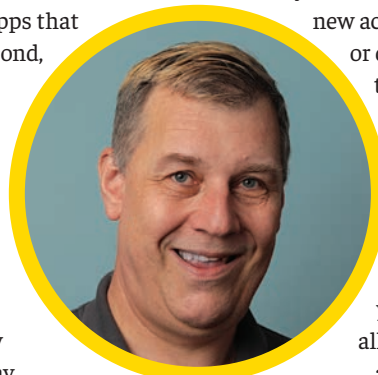
Enable Google Play Protect – Go to the Play Store > Your profile > Play Protect > turn on scanning. Use only

trusted antivirus apps that are highly rated. Stick to the official Play

Store. Many .NET MAUI-based malware spread through unofficial stores or phishing links. Check app permissions – Be wary of apps requesting excessive access, especially if they don't match the app's use.

Stay updated – Security patches close vulnerabilities malware can exploit.

There is not much a user can do to fight system level vulnerabilities, what you can do is try to always keep your operating systems and apps completely updated.



Righard Zwienenberg
Senior Research Fellow at ESET



Lukas Stefanko
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Q Are anti-malware apps effective?

A Righard: HVNC keyloggers are designed to evade detection and may slip past traditional antivirus tools. Clues include phone overheating, sudden battery drain, or random crashes and restarts. To protect against credentials, enable Multi-Factor Authentication (MFA). Even if your login details and an MFA code are stolen, time-based MFA will likely expire before attackers can act, rendering the stolen credentials useless.

Q What to do if data is leaked to the Dark Web?

A Righard: Once sensitive data is compromised, vigilance is crucial. Watch closely for unauthorised activity, new accounts, purchases, or delivery confirmations. Criminals may use your identity to place orders to your address, then intercept or reroute it. Upon learning of a breach, you should update all online passwords and verify that your contact information and delivery addresses haven't been altered.

Stay alert for signs of misuse across all your accounts.

Q What are the roles of data backups in malware attacks?

A Righard: Backups are your best defence – make them frequently and redundantly. Maintain both a local backup (via an external drive or disk image) and a cloud-based one. Crucially, disconnect the backup source after completion to avoid contamination during an attack. If you suspect some compromise, don't rush to restore; doing so may corrupt your clean backup. Instead, consult your security provider for support. Remember: one careless restore can compromise the entire safety net backup. **Q**